

IT Deployment Guide

ULA Claims Hub - Internal Windows VM

Production URL: <https://ula.company.local>

OBJECTIVE

Host the application permanently on an internal VM, provide one trusted HTTPS address to all LAN/VPN users, keep Node.js and application storage private, and ensure automatic restart and recoverable backups.

Target request path



IT-provided infrastructure

- Windows Server VM with a static IP, 4 vCPU, 8 GB RAM minimum, expandable disk, and LAN/VPN-only access.
- Internal DNS A record: ula.company.local -> VM static IP. Use this name only if IT owns and manages the company.local DNS zone.
- Internally trusted TLS certificate whose Subject Alternative Name includes ula.company.local.
- IIS with URL Rewrite and Application Request Routing (ARR), bound to HTTPS port 443.
- Dedicated non-administrator service account, automatic service recovery, monitoring, and nightly backups.

Network exposure

ALLOWED FROM LAN/VPN
TCP 443 to the VM

NOT EXPOSED
TCP 8787; bind Node to 127.0.0.1

CANONICAL URL

APP_BASE_URL must be exactly <https://ula.company.local> - no localhost, port number, or trailing slash.

1. Install and build the application

Recommended application location: C:\Apps\ULA. Install an IT-approved current Node.js LTS release, then run:

```
cd C:\Apps\ULA
npm ci
npm run build
npm start
```

2. Production environment

```
NODE_ENV=production
HOST=127.0.0.1
PORT=8787
APP_BASE_URL=https://ula.company.local
DATABASE_URL=postgres://ula_app:<password>@db-host:5432/ula
DATABASE_SSL=true
DATABASE_RUNTIME_ROLE=ula_app
VITE_SQL_BACKEND=true
```

- Create PostgreSQL with separate migration-owner and runtime roles. The runtime role must not own the schema, be superuser, or have BYPASSRLS.
- Store .env only on the VM. Do not send .env, secrets, .data, or local state to IT.
- Keep all email, AI-provider, and API credentials server-side; never prefix secrets with VITE_.
- Build after VITE_SQL_BACKEND=true is set, then restart the service after changing .env.

3. PostgreSQL setup and migration

```
create database ula;
create role ula_migrator login password '<migration-secret>';
create role ula_app login password '<runtime-secret>';
alter database ula owner to ula_migrator;
set DATABASE_URL and DATABASE_SSL=true
npm run db:migrate
npm run db:check
```

- Run migration and grants with the migration owner. DATABASE_RUNTIME_ROLE defaults to ula_app.
- The connectivity check must return ok:true before the application service is started.
- PostgreSQL stores authentication, claims, documents, leave, employee, and audit data in production.

4. Configure IIS

- 1 Install IIS, URL Rewrite, and ARR; enable ARR proxy functionality.
- 2 Create the ula.company.local site and bind HTTPS port 443 to its matching internal certificate.
- 3 Proxy every request path to http://127.0.0.1:8787 while preserving the original host and forwarding the HTTPS scheme.
- 4 Redirect HTTP port 80 to HTTPS, disable directory browsing, and configure request-size limits to accommodate claim uploads.
- 5 Allow inbound TCP 443 from approved LAN/VPN ranges. Do not create an inbound firewall rule for 8787.

5. Run Node as a Windows service

Production security gate

MUST COMPLETE BEFORE GO-LIVE

Production uses PostgreSQL and server-authoritative authentication. Local JSON state, quick-login personas, seeded temporary credentials, and direct authentication-database synchronization are disabled for production.

- Confirm quick/demo login and seeded temporary credentials are absent from the production build and database.
- Never expose password hashes or permit unauthenticated replacement of authentication state.
- Enforce administrator authorization for account creation, role changes, access revocation, and password reset.
- Enable security headers, login/audit logging, session expiry, and least-privilege filesystem permissions.

Persistence and backup

Critical path: PostgreSQL database plus the uploads directory

- Back up PostgreSQL nightly with `pg_dump` in custom format; retain encrypted off-host copies and multiple restore points.
- Back up the uploads directory separately. Do not back up or transmit `.env`, `secrets`, `node_modules`, `dist`, or `.data` as application state.
- Restore with `pg_restore` into a clean database, run `npm run db:migrate`, restore uploads, and perform the connectivity and health checks.
- Perform and document a restore test on a staging VM. Store secrets in the approved IT secret manager and reissue them during recovery.

Acceptance checklist

- ☐ DNS resolves `ula.company.local` to the VM.
- ☐ HTTPS certificate is trusted with no browser warning.
- ☐ `http://127.0.0.1:8787/api/health` returns `{"ok":true}` on the VM.
- ☐ `npm run db:check` returns `ok:true` and identifies the expected database and runtime user.
- ☐ Port 443 works from LAN/VPN; port 8787 is blocked remotely.
- ☐ Two workstations see the same employee, claim, and uploaded document data.
- ☐ Email links open `https://ula.company.local`.
- ☐ The app returns automatically after a VM reboot.
- ☐ A backup has been restored successfully in a test location.

Useful verification commands

```
# On the VM
Invoke-WebRequest http://127.0.0.1:8787/api/health
# From an employee workstation
Resolve-DnsName ula.company.local
Test-NetConnection ula.company.local -Port 443
```

MICROSOFT REFERENCES

[DNS resource records](#) | [AD Certificate Services](#) | [IIS ARR reverse proxy](#)